AEO “Nazarbayev Intellectual Schools”

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**CHEMISTRY** **Grade 12**

Paper 2 **May 2025**

**2 hours**

Candidates answer on the Question Paper.

Additional Materials: Calculator, Ruler, Data Booklet **12CHEM/02**

**READ THESE INSTRUCTIONS FIRST**

Write your centre number and candidate number in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue, or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 100.

Answer all questions in English.

| For Examiner's Use | |
|---|---|
| **1** | |
| **2** | |
| **3** | |
| **4** | |
| **5** | |
| **6** | |
| **7** | |
| **8** | |
| **Total** | |

This document consists of **20** printed pages.

NIS_G12_CHE_02/2RP/AFP

For Examiner's Use

1 Chemical elements are arranged according to their atomic number in the Periodic Table. In the Periodic Table, the properties of elements are described based on their position, moreover, various trends in properties can be predicted across the periods and down the group.

**(a)** This question is about the elements in Period 3.

State and explain the general trend in atomic radius of the elements across Period 3.

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.................................................................................................................................... [3]

**(b)** **(i)** Arrange the cations $Na^{+}$, $Mg^{2+}$, $Al^{3+}$ in increasing order of ionic radius.

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**(ii)** Explain the difference in size between a magnesium atom and a magnesium ion.

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**(c)** Each element has a value for its first ionisation energy. This value is given in the *Data Booklet*.

**(i)** Define the term *first ionisation energy.*

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**(ii)** Write an equation to represent the first ionisation energy of phosphorus.

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**(iii)** Fig. 1 represents the first ionisation energy values of elements across Period 3.

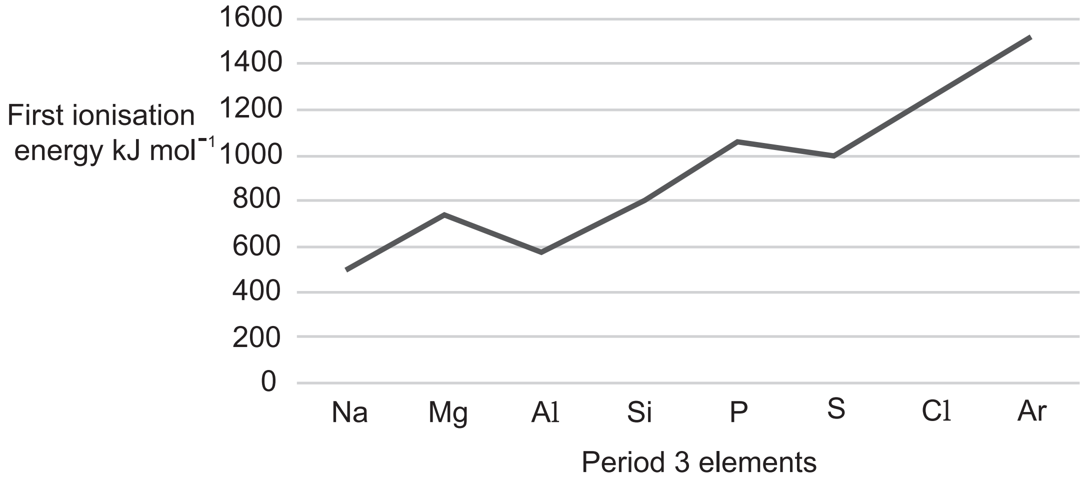


**Fig. 1**

Draw a cross (x) on **Fig. 1** to show the second ionisation energy of phosphorus.

[1]

**(iv)** Explain the difference between the first and second ionisation energies of phosphorus.

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.................................................................................................................... [1]

**[Total: 11]**

For Examiner's Use

**2** The dissociation of water is a reversible reaction.

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq) \qquad \Delta H = +58\ kJ\ mol^{-1}$$

**(a)** State **two** characteristics of a reversible reaction at equilibrium.

**1** ..............................................................................................................................

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**2** ..............................................................................................................................

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[2]

**(b)** **(i)** Write an expression for the ionic product of water, $K_w$.

Include the unit for the ionic product of water.

[2]

**(ii)** State and explain how an increase in temperature affects the value of the ionic product of water, $K_w$.

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**(c)** $K_w$ is also used to calculate the pH of various solutions.

**(i)** Use the value of $K_w$ from the *Data Booklet* to calculate the pH of 0.010 mol $dm^{-3}$ KOH.

pH = .................................... [2]

For Examiner's Use

**(ii)** Methylamine is a weak base, which dissociates partially in water.

$$CH_3NH_2(aq) + H_2O(l) \rightleftharpoons CH_3NH_3^+(aq) + OH^-(aq)$$

Write down the expression for $K_b$ for this reaction.

[1]

**(iii)** $K_b$ for this reaction is $4.38 \times 10^{-4}$ mol dm$^{-3}$. Calculate the pH of 0.010 mol dm$^{-3}$ $CH_3NH_2$.

pH = .......................................... [4]

**(iv)** Explain the difference between pH values calculated in **c (i)** and **c (iii).**

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**[Total: 14]**

**3** In ethanolic solution, 2-bromo-2-methylpropane, $(CH_3)_3CBr$, reacts with potassium cyanide, KCN. This is a nucleophilic substitution reaction.

For Examiner's Use

$$(CH_3)_3CBr + KCN \rightarrow (CH_3)_3C\text{-}C\equiv N + KBr$$

**(a)** State what is meant by the term *nucleophile.*

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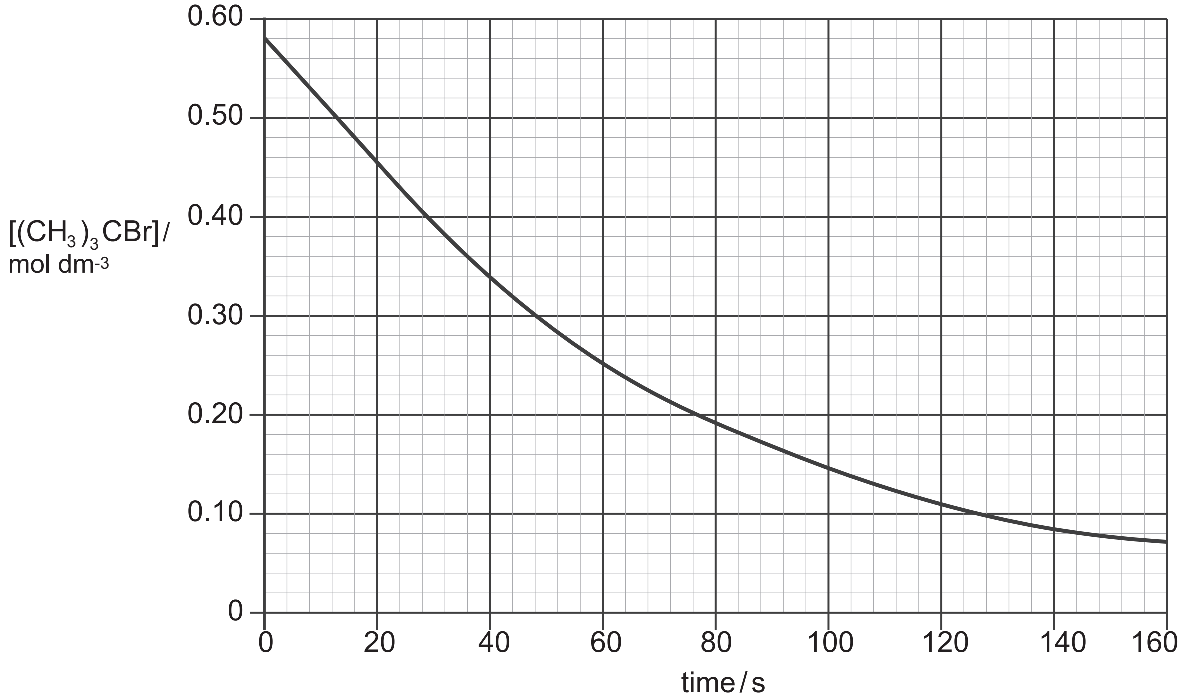
**(b)** Show the mechanism for this reaction. Include all necessary curly arrows, lone pairs, and relevant dipoles.

[3]

**(c)** The kinetics of this reaction is studied and the graph of the concentration of a sample of 2-bromo-2-methylpropane plotted against time is shown below. The rate of this reaction is analysed using potassium cyanide in a large excess.



For Examiner's Use

**(i)** Draw a tangent to the curve when the concentration of 2-bromo-2-methylpropane is 0.30 mol $dm^{-3}$.

[1]

**(ii)** The initial rate of the reaction in this experiment has the value $8.81 \times 10^{-3}$ mol $dm^{-3}$ $s^{-1}$. Use this value and a rate obtained from your tangent, to show that the order of the reaction is 1st with respect to 2-bromo-2-methylpropane.

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**(iii)** Write the rate equation for the reaction if the order of the reaction with respect to KCN is zero.

[1]

**(iv)** Calculate the value for the rate constant, k, and deduce its units.

k = ............................... units ............................ [1]

**(d)** In another experiment 2-iodo-2-methylpropane is used instead of 2-bromo-2-methylpropane. 2-iodo-2-methylpropane reacts faster with KCN than 2-bromo-2-methylpropane**.** Explain the reason.

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**[Total: 13]**

For Examiner's Use

**4** Proteins are formed by the polymerisation of amino acids.

**(a) (i)** State the type of polymerisation used to form proteins.

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**(ii)** The amino acids alanine and valine can be joined together to form a dipeptide.

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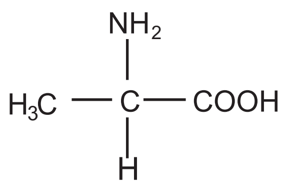
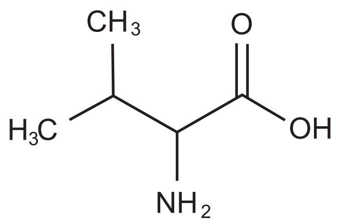
|---|---|
|  |  |
| alanine, *ala* | valine, *val* |

Draw the structure of the dipeptide ala-val.

[1]

**(iii)** Draw the zwitterion of valine.

[1]

**(iv)** Draw the species formed by alanine at high pH.

[1]

For Examiner's Use

**(b)** Proteins can form tertiary structure. Different forces and bonds stabilise the tertiary structure.

List **four** types of side-chain interactions and state the groups that are responsible for the formation of these interactions between amino acids residues.

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**(c)** Valine exists as a pair of stereoisomers.

**(i)** Explain the meaning of the term *stereoisomers.*

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**(ii)** Explain how the pair of stereoisomers of valine can be distinguished.

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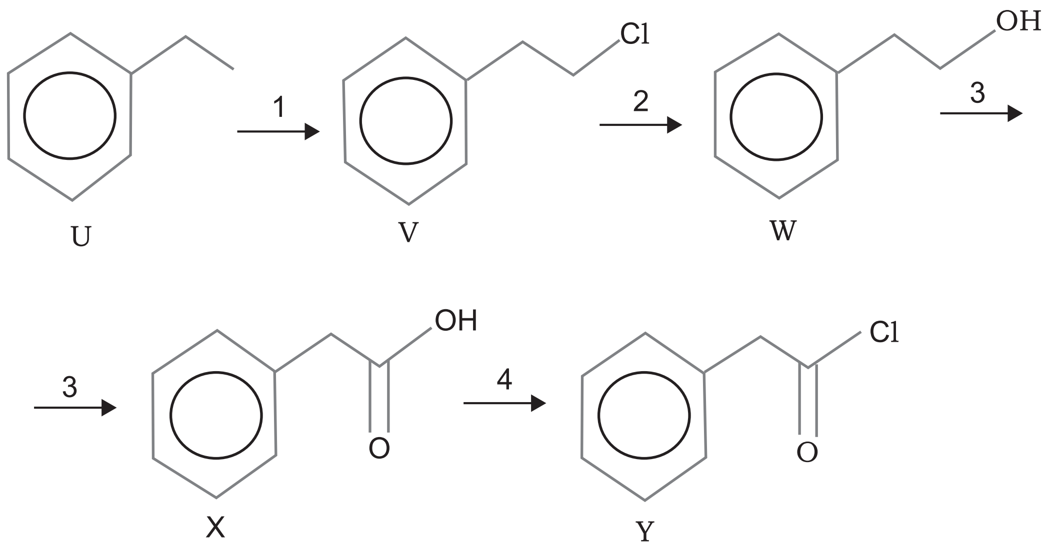
**[Total: 11]**

5 Phenylacetyl chloride, compound **Y**, is a colourless volatile liquid. It is used in manufacturing esters for flavourings. It can be synthesised from ethylbenzene, **U**, by the given route.

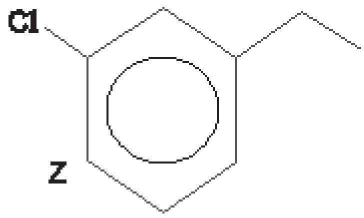





**(a)** Suggest reagents and conditions for Reaction 1.

........................................................................................................................ [1]

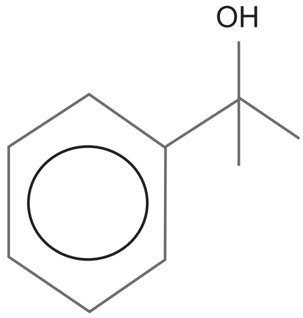
**(b)** Give reagent and catalyst to produce compound **Z** (an isomer of **V**) from ethylbenzene.




Z

........................................................................................................................ [1]

**(c)** Suggest a test that can help to distinguish compound **W** from compound **A**.




A

Reagents

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Observation with **W**

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Observation with **A**

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[3]

**(d)** Give the name of the mechanism for Reaction 2.

..................................................................................................................................... [1]

**(e)** In order to ensure that the oxidation to phenylacetic acid, compound **X**, is complete, the reaction is carried out under reflux.

Describe what happens when a reaction mixture is refluxed and why it is necessary, in this case, for a complete oxidation to phenylacetic acid.

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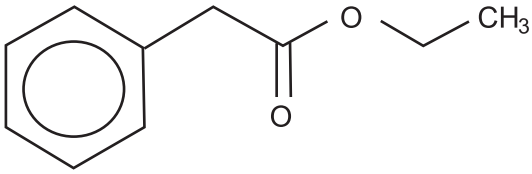
For Examiner's Use

**(f)** Phenylacetyl chloride, compound **Y**, reacts with ethanol to produce ester **B**.



**B**

**(i)** Name the mechanism for the reaction between phenylacetyl chloride and ethanol.

.................................................................................................................... [1]

**(ii)** Outline a mechanism for this reaction. You can represent aryl group as "R".
Include all necessary curly arrows, lone pairs and relevant dipoles.

[4]

**[Total: 13]**

For Examiner's Use

**6** In this question, give all your answers in **three** significant figures.

**(a)** Zinc nitrate decomposes on heating to form zinc oxide, nitrogen (IV) oxide and oxygen.

**(i)** Write a balanced chemical equation for this decomposition reaction. Include state symbols.

.......................................................................................................................... [2]

**(ii)** Thermal decomposition of a sample of zinc nitrate produced 2.16 g of zinc oxide.

Calculate the amount, in moles, of zinc oxide in 2.16 g.

amount = ................................ mol [1]

**(iii)** Calculate the total amount, in moles, of gas produced from this sample of zinc nitrate.

amount = ................................ mol [1]

**(b)** In another experiment, a different sample of zinc nitrate decomposed to produce 0.506 mol of gas. Calculate the volume, in $dm^3$, that this gas would occupy at 357 K and $1.00 \times 10^5$ Pa (The gas constant R = 8.31 J $K^{-1}$ $mol^{-1}$).

volume = ................................ $dm^3$ [2]

For Examiner's Use

**(c)** 0.0247 mol sample of zinc oxide, produced from the decomposition of zinc nitrate, was reacted with nitric acid.

**(i)** Write a balanced chemical equation for this reaction.

.............................................................................................................................. [1]

**(ii)** Calculate the amount, in moles, of acid needed to react completely with the 0.0247 mol sample of zinc oxide.

amount = ................................ mol [1]

**(iii)** 0.0247 mol sample of zinc oxide needed 45.7 $cm^3$ of nitric acid for complete reaction. Use this information and your answer to part **(c)(ii)** to calculate the concentration, in mol $dm^{-3}$, of the nitric acid.

amount = .......................................... mol $dm^{-3}$ [1]

**[Total:9]**

7 Isomers of butanol have comparatively better fuel properties and are less corrosive compared to ethanol. One of the isomers of butanol, **X**, is analysed by IR and $^{1}$H-NMR spectroscopy and data is collected for further experiments.

For Examiner's Use

**(a)** Suggest how IR spectrum helps the student to determine presence of alcohol group.

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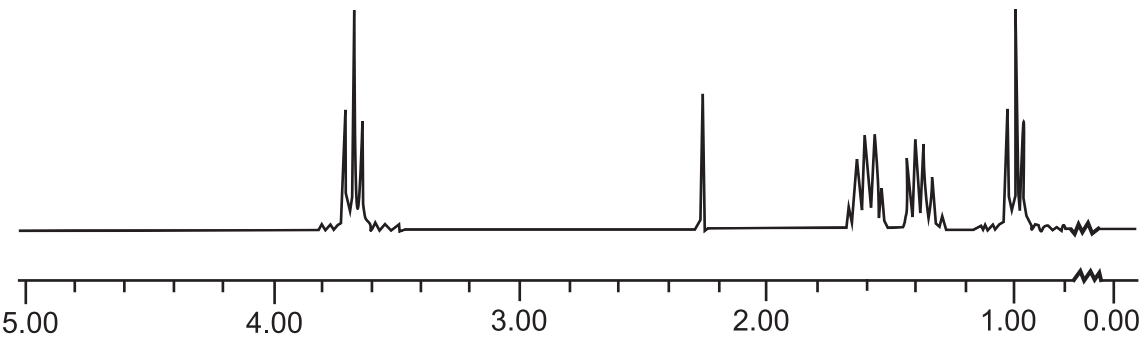
**(b)** The $^{1}$H-NMR spectrum of isomer **X** is shown below.




**(i)** Deduce the number of different types of protons present in a compound.

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**(ii)** The spectrum shows two triplets with different chemical shifts. Identify groups that are responsible for each chemical shift.

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**(iii)** Explain the reason why peak at $\delta = 3.63$ shifts more to the left.

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**(iv)** Deduce the structural formula of an isomer **X.**

[1]

**(c)** The isomer **X** produces more energy than ethanol when burnt. To demonstrate this, the following experiment is conducted.

**(i)** Write a balanced symbol equation for the complete combustion of the isomer **X**.

............................................................................................................................ [1]

**(ii)** The enthalpy change of combustion is determined. Isomer **X** is placed in a spirit burner and weighed. The burner is then lit and allowed to burn for a few minutes. The heat produced by combustion is used to warm 50 g of water in a copper calorimeter.

The burner is then re-weighed. The temperature of water is recorded before and after heating.

The results obtained are represented in **Table 1.**

**Table 1**

| | |
|---|---|
| Initial mass of spirit burner and isomer **X** / g | 205.79 |
| Final mass of spirit burner and isomer **X** / g | 205.60 |
| Initial temperature of water / °C | 24.5 |
| Final temperature of water / °C | 39.8 |

Use results from the table to calculate a value, in kJ mol$^{-1}$, for the enthalpy change when one mole of isomer **X** is burned.
(The specific heat capacity of water is 4.18 J K$^{-1}$ g$^{-1}$)

[3]

**(iii)** The results from **(c)(ii)** are less than the standard values for the enthalpy change of combustion of butanol.

Suggest **two** reasons why the obtained results from part **c (ii)** are less.

1 ..........................................................................................................................

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2 ..........................................................................................................................

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[2]

For Examiner's Use

**(d)** Bond energies can also be used to calculate the standard enthalpy of combustion of a compound.

**(i)** Use the *Data Booklet* to calculate a value for the standard enthalpy of combustion of isomer **X**. Show your working.

[3]

**(ii)** The standard enthalpy of combustion of isomer **X** is -2670 kJ $mol^{-1}$. Explain why this value differs from the value obtained using bond energies.

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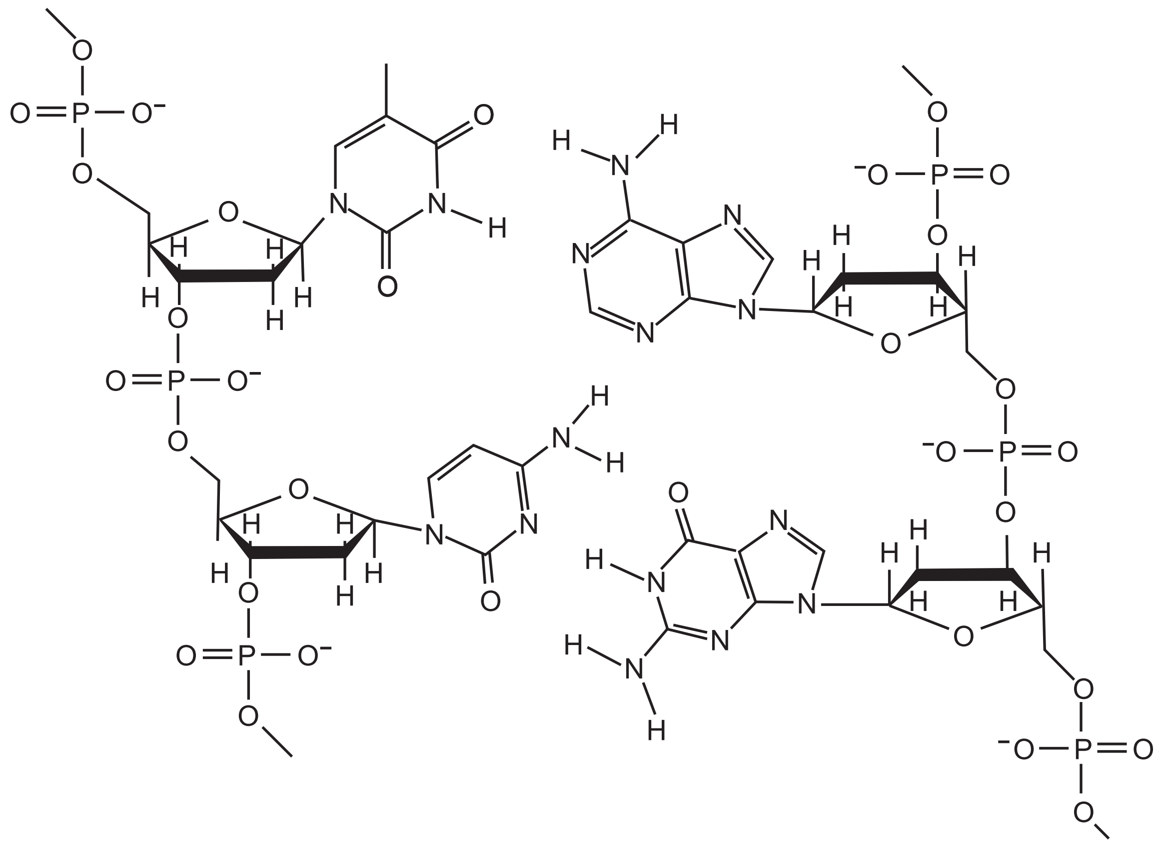
**[Total: 16]**

**8** A single strand of DNA is a polymer – it is made up of four different monomers, nucleotides bonded together. These monomers are called nitrogenous bases.



**(a)** DNA exists as two strands of nucleotides in the form of a double helix with hydrogen bonding between the strands.

**(i)** Use these structures of nitrogenous bases to show hydrogen bonds between base pairs.

[2]

**(ii)** Name the correct complementary base pairing for DNA.

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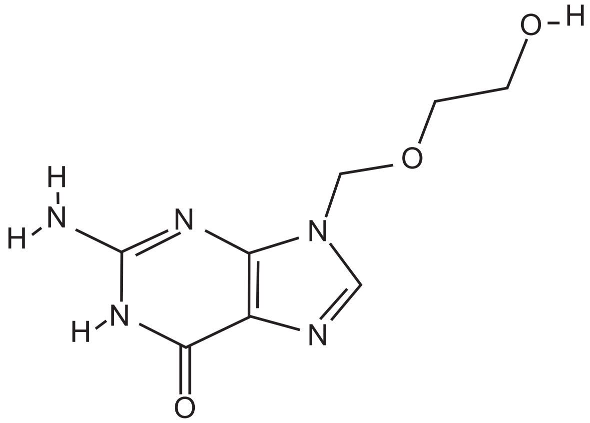
.......................................................................................................................... [1]

*For Examiner's Use*

**(b)** Acyclovir tricks the viral enzymes into using it as building blocks of DNA and thus prevents the virus from multiplying.

The structure of *Acyclovir*



**(i)** Indicate the amide bond in the structure of the acyclovir molecule.

[1]

**(ii)** Suggest why it is better to take medicine by injection rather than by mouth.

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**(iii)** Describe what side effects may occur when taking medications and how to eliminate them.

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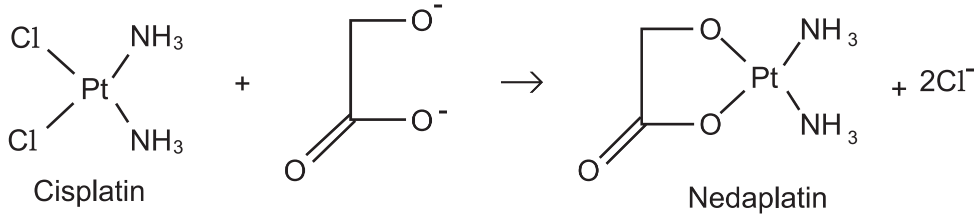
**(c)** *Nedaplatin*, a new anticancer drug with less side effects, has been developed.

It can be synthesised from *cis*-platin by the following reaction.




**(i)** Explain why the reaction occurs in terms of the chelate effect.

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**(ii)** In recent years, scientists have developed a variety of different drug delivery systems with less side effects. They found out that gold nano particles can be effectively used in cancer treatment.

Suggest how gold nano-cages selectively deliver cancer-fighting drugs to the sites of tumours in the body.

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**(d)** *Cis*-platin has a geometrical isomer.

**(i)** Draw 3D structure of its isomer – *trans*-platin.

[1]

**(ii)** Explain why *trans*-platin is not used as anticancer drug.

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.................................................................................................................... [1]

**[Total: 13]**

For Examiner's Use